The Shark & Circle Problem

A Complete Mathematical Solution

Classical pursuit problem — optimal escape strategy & critical bound

1 The Problem

A swimmer starts at the **exact centre** of a circular pond of radius R. A shark sits on the shore and can swim along the perimeter at k times the swimmer's speed. Both move optimally. **Can the swimmer escape?**

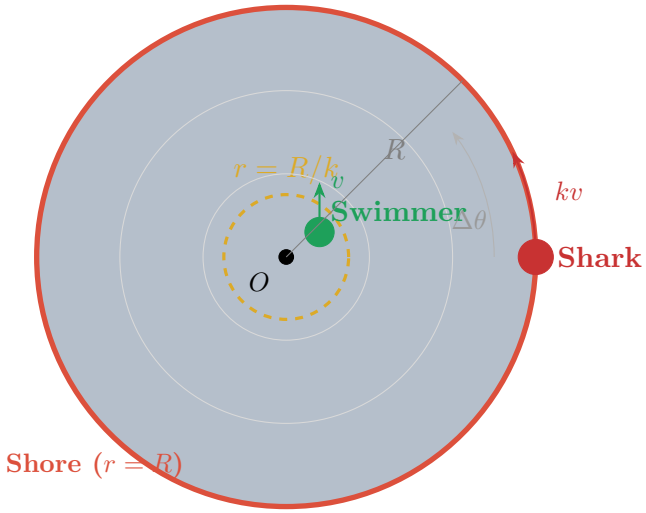


Figure 1: Setup: swimmer at centre, shark on the shore, critical orbit shown dashed.

2 Assumptions

- The pond is a perfect circle of radius R.
- The swimmer starts at the origin with speed $v_s = 1$.
- The shark is confined to the shore and moves at speed $v_{sh} = kv_s = k$.
- The shark always moves to minimise the angular gap $\Delta\theta$ (optimal pursuit).
- The swimmer is **caught** if the shark is at the swimmer's shore landing point at the moment of arrival.
- Both parties have full information — they can see each other at all times.
- No boundaries other than the shore; no currents; continuous time.

3 Key Quantities

Let the swimmer's position be (r, θ) in polar coordinates centred at O. Let ϕ denote the shark's angular position on the shore.

Symbol	Meaning	Value / Units
R	Pond radius	10 m (example)
v	Swimmer speed	1 m/s
k	Shark speed ratio	$k > 1$
$r(t)$	Swimmer's radial distance	$r \in [0, R]$
$\theta(t)$	Swimmer's angle	$\theta \in [0, 2\pi)$
$\phi(t)$	Shark's angle on shore	$\phi \in [0, 2\pi)$
$\Delta\theta$	Angular gap: $\theta - \phi$	signed, wrapped to $(-\pi, \pi]$
ω_s	Swimmer's angular velocity	$\dot{\theta} = v \sin\alpha / r$
ω_{sh}	Shark's max angular velocity	kv/R

4 The Critical Radius

Lemma 1 (Angular Velocity Advantage)**.** When the swimmer is at radial distance $r < R/k$, they can orbit the centre strictly faster than the shark can circle the shore, regardless of the direction chosen.

Proof. The swimmer's maximum angular velocity at radius r is:

$$\omega_s^{\max} = \frac{v}{r}$$

The shark's angular velocity on the shore is bounded by:

$$\omega_{sh}^{\max} = \frac{kv}{R}$$

For the swimmer to orbit faster:

$$\frac{v}{r} > \frac{kv}{R} \quad \Longleftrightarrow \quad r < \frac{R}{k}$$

Inside this radius, $\omega_s > \omega_{sh}$, so the swimmer can maintain any desired angular gap, including widening it to π (diametrically opposite to the shark). □ □

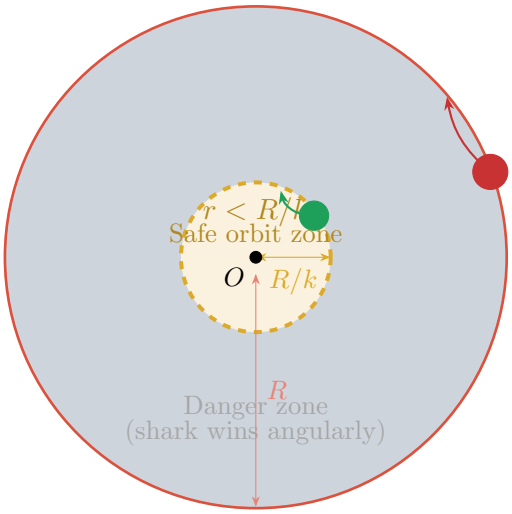


Figure 2: The critical radius R/k. Inside it, the swimmer orbits faster than the shark.

5 The Optimal Two-Phase Strategy

Phase 1 — Orbit to Gain Angular Lead

The swimmer moves to just inside the critical radius $r^* = R/k - \varepsilon$, then orbits in the direction that *increases* $|\Delta\theta|$. Because $\omega_s > \omega_{sh}$ at this radius, the angular gap grows monotonically. The swimmer continues orbiting until:

$$|\Delta\theta| = \pi \quad \text{(shark is diametrically opposite)}$$

The angular rate of gain is:

$$\dot{(\Delta\theta)} = \omega_s - \omega_{sh} = \frac{v}{r^*} - \frac{kv}{R}$$

Time to reach $\Delta\theta = \pi$ from zero:

$$T_1 = \frac{\pi}{\dfrac{v}{r^*} - \dfrac{kv}{R}} = \frac{\pi R r^*}{v(R - kr^*)}$$

Phase 2 — Radial Sprint to Shore

Once $|\Delta\theta| = \pi$, the swimmer sprints **radially outward** toward the nearest shore point. The swimmer has angular head-start π on the shark.

Time to reach shore from r^*:

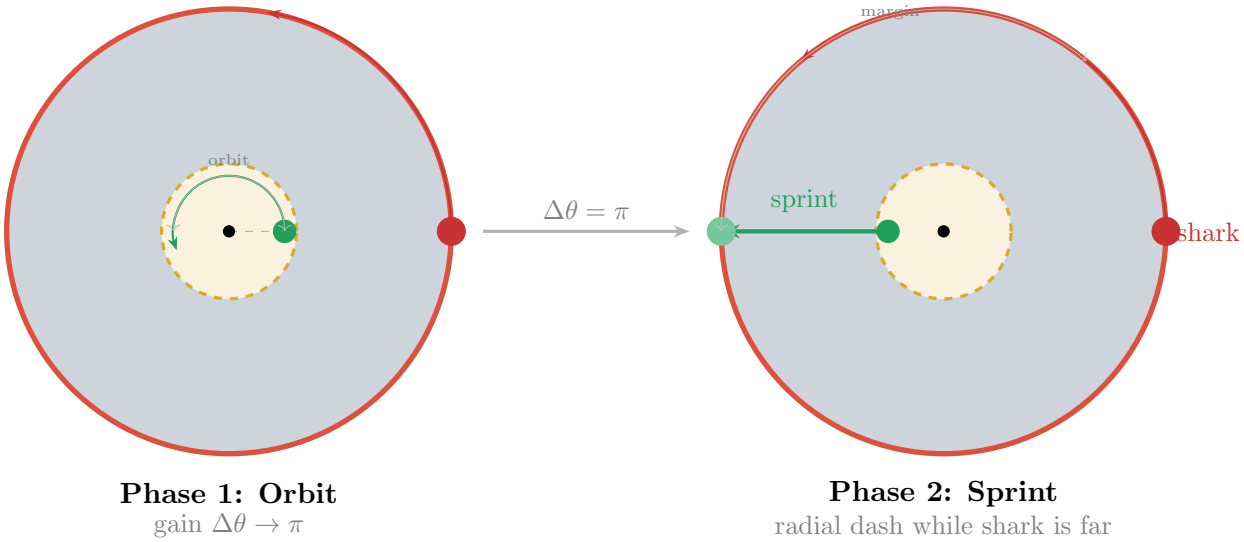
$$T_2 = \frac{R - r^*}{v}$$

In this time, the shark traverses an arc of angle:

$$\Delta\phi = \frac{kv}{R} \cdot T_2 = \frac{k(R - r^*)}{R}$$

The swimmer arrives at angle θ_0; the shark arrives at $\theta_0 + \pi - \Delta\phi$ (measuring from the landing point). The **escape margin** in radians is:

$$\boxed{\delta_{\text{margin}} = \pi - \Delta\phi = \pi - \frac{k(R - r^*)}{R}}$$



Phase 1: Orbit
gain $\Delta\theta \to \pi$

Phase 2: Sprint
radial dash while shark is far

Figure 3: Left: Phase 1 — orbit inside R/k to reach a 180 lead. Right: Phase 2 — sprint radially while the shark is opposite.

6 The Critical Bound

For escape to be possible, the margin must be positive:

$$\delta_{\text{margin}} > 0 \quad \Longrightarrow \quad \pi - \frac{k(R - r^*)}{R} > 0 \quad \Longrightarrow \quad k < \frac{\pi R}{R - r^*}$$

Taking $r^* \to 0$ (swimmer starts sprinting from just inside the critical circle and we optimise):

$$\lim_{r^* \to 0^+} \frac{\pi R}{R - r^*} = \pi$$

But wait — if we use the optimal $r^* = R/k - \varepsilon$, substituting:

$$\delta_{\text{margin}} = \pi - \frac{k\left(R - \frac{R}{k}\right)}{R} = \pi - \frac{k \cdot R(1 - \frac{1}{k})}{R} = \pi - (k - 1)$$

Escape requires $\delta_{\text{margin}} > 0$:

$$\pi - (k - 1) > 0 \quad \Longrightarrow \quad k < \pi + 1$$

Theorem 1 (Critical Speed Ratio)**.** The swimmer can escape if and only if:

$$k < \pi + 1 \approx 4.1416$$

When $k = \pi + 1$, escape is exactly marginal (zero-margin). When $k > \pi + 1$, no strategy guarantees escape.

7 Escape Margin in Metres

At the optimal sprint radius $r^* = R/k$, substituting into the margin formula:

$$\delta_{\text{margin}} = \pi - (k - 1) \quad \text{[radians]}$$

Converting to **arc length** (distance along shore between swimmer and shark at landing):

$$\boxed{d_{\text{escape}} = R \cdot \big[\pi - (k - 1)\big] \quad \text{[metres]}}$$

k	$\pi - (k-1)$ (rad)	d_{escape} for $R = 10$m	Escapes?
3.00	1.14 rad	11.4 m	**Yes**
3.60	0.54 rad	5.4 m	**Yes**
4.00	0.14 rad	1.4 m	**Yes**
4.14	0.00 rad	0.0 m	**Marginal**
4.25	−0.11 rad	−1.1 m	**No**

Table 1: Theoretical escape margin at various shark speeds, $R = 10$ m.

8 Formal Escape Condition

Let ϕ_{land} be the swimmer's shore landing angle and ϕ_{shark} be the shark's angle at that moment. Define the capture threshold δ_c (angular half-width of the shark's reach).

The swimmer escapes if and only if:

$$|\phi_{\text{land}} - \phi_{\text{shark}}| > \delta_c \qquad (\text{mod } 2\pi)$$

In the two-phase strategy, this becomes the condition:

$$\pi - (k - 1) > \delta_c \quad \Longrightarrow \quad k < \pi + 1 - \delta_c$$

For any finite capture threshold $\delta_c > 0$, the bound is slightly stricter than $\pi + 1$.

9 Why No Better Strategy Exists for $k \geq \pi + 1$

Upper bound argument. When $k \geq \pi + 1$, the shark can always cover any shore point from any other in the time the swimmer needs to travel from anywhere inside the pond to the shore.

Formally, the swimmer must travel at least distance $R - \varepsilon$ radially (they start at the centre). The fastest this can be done is time $T_{\min} = (R - \varepsilon)/v$. In that time the shark covers arc:

$$kv \cdot T_{\min} = k(R - \varepsilon)$$

For the shark to cover a full semicircle πR (enough to reach any shore point from any starting position):

$$k(R - \varepsilon) \geq \pi R \quad \Longrightarrow \quad k \geq \frac{\pi R}{R - \varepsilon} \to \pi + 1 \quad \text{as} \quad \varepsilon \to 0$$

When $k \geq \pi + 1$, the shark can always intercept — it covers more than a semicircle while the swimmer crosses from the centre region to the shore.

10 Summary

The Shark & Circle — Solution in three lines:

1. Stay inside $r < R/k$ and orbit to gain a 180 angular lead on the shark.
2. Once opposite, sprint radially to the shore in time $T_2 = (R - R/k)/v$.
3. Escape margin: $d = R[\pi - (k - 1)]$ metres. **Escape is possible iff** $k < \pi + 1 \approx 4.14$**.**

This document accompanies the browser simulation at
gouthamchoudary.github.io/shark_and_circle